Demand Forecasting and Menu Optimization for Operational Efficiency in a Cloud Kitchen

A Proposal report for the BDM capstone Project

Submitted by

Name: M.HEMA VARSHNI

Roll number: 24f1001256

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled **"Demand Forecasting and Menu Optimization for Operational Efficiency in a Cloud Kitchen"**. I extend my appreciation to **Padma's Home Food owned by Mrs.Padmavathy Ravichandran**, for providing the necessary resources that enabled me to conduct my project.

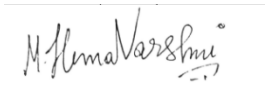
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: 

Name: M.HEMA VARSHNI

Date: 13-11-2025

1 Executive Summary and Title

Padma's Home Food is a cloud kitchen, owned by Mrs.Padmavathy Ravichandran, is operating through the Cookr app platform, in her house, located in Medavakkam Main road, Chennai. Operating on a B2C model within a 2-3 km radius, the business has established strong market presence by Indian home-cooked meals across breakfast, lunch, and dinner categories, offering over 30 dishes through both on-demand orders and monthly subscriptions with 45-minute preparation windows.

Despite operational success, the business faces profitability challenges from unpredictable demand. Festival and holiday surges cause ingredient stockouts, forcing emergency procurement through Zepto and Swiggy with 20-30% surge pricing, GST, delivery fees (₹20-40), and platform fees, the costs avoided through planned local purchasing. These expenses erode thin margins already constrained by quality ingredients and labor costs. Additional challenges include capacity strain causing closures, family dependency for support, and illness-related disruptions.

This project will analyze historical order data to develop a demand forecasting model and operational optimization framework that enables strategic inventory planning, enhanced capacity management, and improved profitability while preserving the quality standards that differentiate Padma's Home Food.

2 Organization Background

Padma's Home Food is a cloud kitchen established in 2024 by Mrs. Padmavathy Ravichandran, operating through the Cookr app platform in Dwaraka Apartments, Medavakkam Main Road, Chennai, Tamil Nadu. This micro-enterprise specializes in authentic home-style Indian cuisine (vegetarian and non-vegetarian), serving customers within a 2-3 km radius that adjusts based on demand and platform availability.

The kitchen offers over 30 dishes across breakfast (idli, dosa, pongal), lunch and dinner (sambar rice, biryani, curd rice), and beverages through a made-to-order model with 45-minute preparation windows. The operational team comprises Mrs. Padmavathy as primary cook, a helper for utensil cleaning and vegetable preparation, and her husband managing packing operations.

The business operates through on-demand orders (same-day or advance) and monthly subscriptions with customized dishes. Mrs. Padmavathy has established strong reputation for authentic taste and quality, generating consistent customer demand and repeat orders.

3 Problem Statement

3.1 Frequent ingredient shortages and costly last-minute quick commerce orders reduce profit margins by 20–30%. A data-driven demand forecasting system predicting order volumes 3–7 days ahead, considering trends, seasons, and events. This enables bulk, planned procurement, reducing emergency costs and ensuring steady supply.

3.2 Sudden peak-hour surges cause cook exhaustion and unplanned closures. By analyzing order patterns, the business can determine optimal operating hours, capacity, and staffing, maximizing efficiency, maintaining service quality, and boosting revenue.

3.3 Some dishes have low margins or high preparation costs. Menu profitability analysis helps identify high-margin items and optimize pricing, improving profit margins while maintaining quality and customer satisfaction.

4 Background of the Problem

Padma's Home Food has gained popularity on the Cookr platform due to its authentic, home-cooked meals. However, rapid growth has exposed operational challenges that threaten profitability. Demand is highly unpredictable, with sudden surges during festivals, national and local holidays, often overwhelming kitchen capacity. Currently, ingredient planning relies on intuition, leading to frequent stockouts during peak periods. Emergency procurement through quick commerce incurs 15–25% higher costs, directly reducing profit margins.

As the sole cook, Padma faces exhaustion during extended peak hours, forcing early lunch or dinner closures and resulting in lost revenue and customer dissatisfaction. Dependence on family for assistance is unreliable, and illness can halt operations for 7–10 days, causing cancellations and attrition.

Financial strain is compounded by the commitment to high-quality ingredients and helper wages, leaving thin margins. Bulk planning from local vendors could reduce ingredient costs by 15–25%, but lack of demand forecasting prevents this. Additionally, menu items vary in profitability, and current operations provide no insight into optimal pricing, staffing, or service hours. Without data-driven insights, scaling sustainably and improving profit margins remains unattainable despite strong customer demand.

5 Problem Solving Approach

This project will employ a data-driven approach to address Padma's Home Food's operational challenges through analysis of historical order data and development of practical decision-support tools.

Phase 1: Data Collection and Preparation
Primary data will be sourced from Padma's Cookr app, including order ID, timestamps, dish names, quantities, order values, customer locations, order type, and preparation times. Structured interviews will supplement this with ingredient costs, preparation times, helper wages, profit margins, procurement costs (wholesale vs. Zepto), and operational hours. Data from the past [specify: 6 months/1 year] will capture seasonal variations, festivals, and events. Data cleaning will standardize dish names, categorize meals, and create derived variables such as order density per hour, day-of-week patterns, and peak identification.

Phase 2: Analytical Approach

1. **Demand Forecasting**
    - **Time Series Analysis:** Identify trends, seasonality, and cyclical patterns using Python (pandas, statsmodels).

- **Peak Analysis:** Determine high-demand hours and days.
- **Event Impact:** Correlate spikes with festivals, holidays and vacation time.
- **Dish-Level Trends:** Analyze demand variation across dishes.
  Justification: Forecasting enables proactive inventory planning and resource allocation.

2. **Operational Capacity Analysis**
   - **Hourly Order Heatmaps:** Visualize peak periods.
   - **Preparation Load Analysis:** Evaluate cumulative cooking requirements.
   - **Threshold Identification:** Determine sustainable order volume per hour.
     Justification: Ensures quality service while optimizing labor and kitchen utilization.
3. **Profitability and Menu Optimization**
   - **Dish-wise Profitability & ABC Analysis:** Categorize dishes by profit and volume.
   - **Ingredient Overlap Analysis:** Identify shared ingredients for bulk procurement.
     Justification: Optimizes menu mix to maximize profit without compromising customer satisfaction.
4. **Cost-Benefit Analysis of Procurement**
   - Compare Zepto vs. local vendor costs and emergency order frequency.
   - Determine minimum stock levels to reduce reliance on emergency purchases.
   - Calculate potential savings from planned procurement.
     Justification: Demonstrates clear ROI for implementing forecasting and inventory planning.

Tools and Technologies: Python (pandas, numpy, matplotlib, seaborn), Excel/Google Sheets dashboards, and forecasting models.

Phase 3: Solution Development

- Demand forecasting model with weekly/monthly projections
- Peak period calendar and optimal operating hours
- Ingredient requirement forecasts
- Menu optimization recommendations

This approach converts intuition-based operations into data-driven decision-making, improving efficiency, profitability, and sustainable growth

6 Expected Timeline

6.1 Work Breakdown Structure

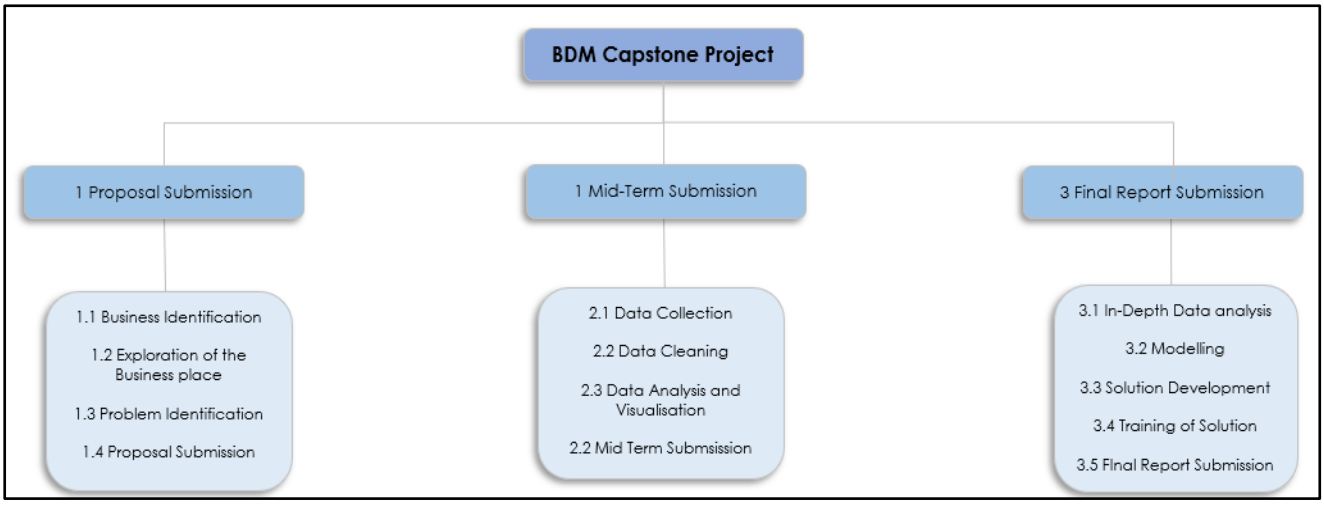
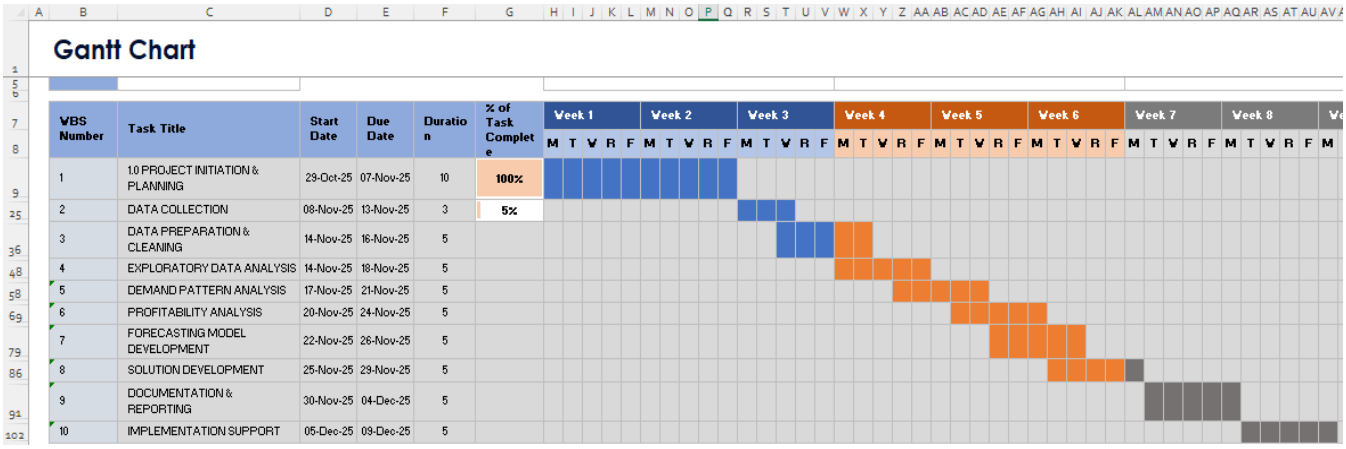


Figure 1 Work Breakdown Structure of the Project

6.2 Gantt chart



Gantt Chart

WBS Number	Task Title	Start Date	Due Date	Duration	% of Task Complete
1	1.0 PROJECT INITIATION & PLANNING	29-Oct-25	07-Nov-25	10	100%
2	DATA COLLECTION	08-Nov-25	13-Nov-25	3	5%
3	DATA PREPARATION & CLEANING	14-Nov-25	16-Nov-25	5	
4	EXPLORATORY DATA ANALYSIS	14-Nov-25	18-Nov-25	5	
5	DEMAND PATTERN ANALYSIS	17-Nov-25	21-Nov-25	5	
6	PROFITABILITY ANALYSIS	20-Nov-25	24-Nov-25	5	
7	FORECASTING MODEL DEVELOPMENT	22-Nov-25	26-Nov-25	5	
8	SOLUTION DEVELOPMENT	25-Nov-25	29-Nov-25	5	
9	DOCUMENTATION & REPORTING	30-Nov-25	04-Dec-25	5	
10	IMPLEMENTATION SUPPORT	05-Dec-25	09-Dec-25	5	

Figure 2(a) Expected timeline for completion of project.

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Figure 2(b) Expected timeline for completion of project.

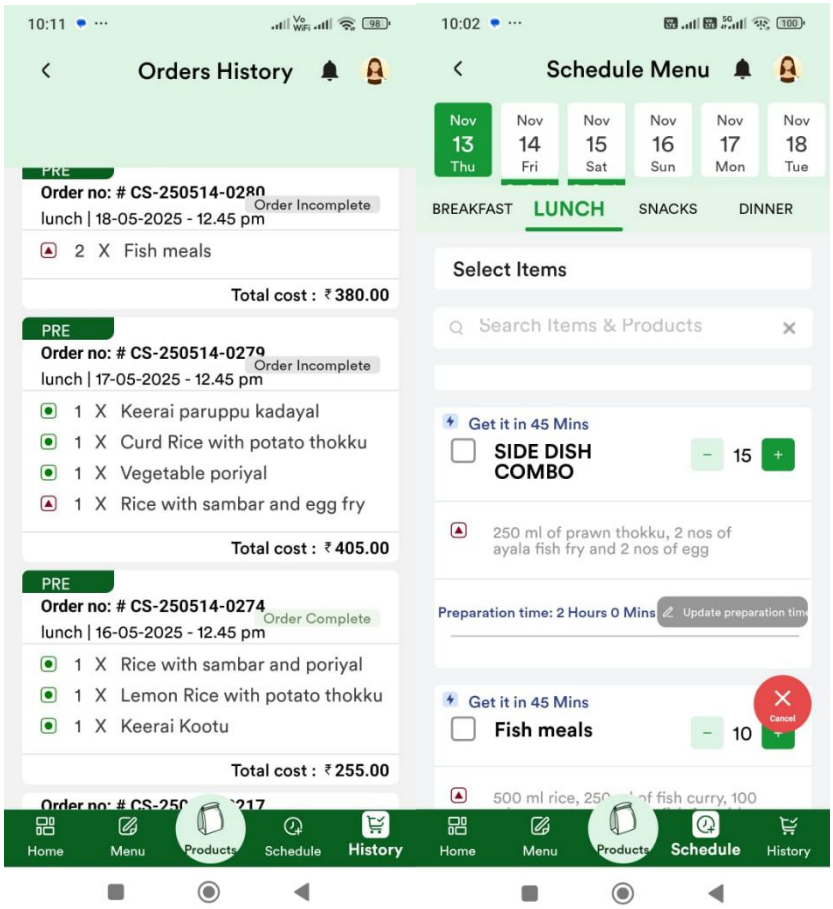

Figure 3 App Layout

7 Expected Outcome

This project will equip Padma's Home Food with data-driven tools and insights to streamline operations and boost profitability.

- A **Demand Forecasting System** will predict high-demand periods (festivals, cricket matches, holidays) 3–7 days in advance, reducing emergency Zepto/Swiggy orders by **70–80%** and saving **₹18,000–30,000 annually**.
- **Procurement Optimization** will compare vendor prices and define reorder points, supported by a weekly checklist to ensure timely, cost-effective purchases.
- **Capacity Management Insights** will guide optimal order volumes and peak-time decisions to prevent burnout while maximizing revenue.
- The **Menu Optimization Framework** will identify profitable dishes, improve pricing, and enhance margins by **12–18%**.